

ASHUTOSH ROY

Data Engineer | Python Backend Specialist

+91-9810599837 | ashu2003roy@gmail.com | Portfolio | Github | Gurugram, Haryana

SUMMARY

Data Engineer and former founder with expertise in building scalable ETL/ELT pipelines and semantic data architectures. Skilled in designing multidimensional data models (Star/Snowflake), optimizing SQL performance, and deploying production backend systems via FastAPI. Experienced in extracting structured insights from diverse datasets using OCR and hybrid retrieval systems (FAISS/Pinecone/Solr). Strong foundation in Python and a deep interest in data ontology, knowledge graphs, and enterprise-grade cloud solutions.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Engineered high-performance **ETL pipelines** for multimodal archaeological datasets; automated document ingestion, semantic extraction (OCR), and structured markdown generation.
- Designed a **hybrid retrieval architecture** combining dense (FAISS/Pinecone) and sparse (Apache Solr) indices, enabling semantic search and entity resolution across thousands of historical records.
- Deployed scalable **FastAPI** backend services for real-time inference, ensuring low-latency data access and sub-second retrieval responses for downstream AI agents.
- Co-authored research on multimodal search pipelines for archival digitization — submitted for peer review at top-tier venues (arxiv.org/abs/2511.05667).

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Built end-to-end **Python data pipelines**: data ingestion → cleaning → transformer-based summarization (Mistral/LLaMA) → API deployment for healthcare applications.
- Architected intent-routing systems and medical record ontologies to standardize summaries, reducing patient query resolution time by ~**40%** through automated data flow.
- Implemented robust OCR and data validation modules to extract structured clinical data from messy image/PDF reports with 90%+ accuracy.

Software Engineer Intern

Nov 2024 – Dec 2024

LeanTactics Solutions Pvt. Ltd.

Remote

- Engineered high-performance backend services using Python and **FastAPI**, optimizing data flow between inference layers and production application systems.
- Achieved a **25–35% reduction** in response latency under production load by profiling API performance bottlenecks and optimizing SQL/NoSQL data interactions.

PROJECTS

Shopping Assistant Agent (LangGraph & MCP) | Python, LangGraph, MCP, API Integration

Feb 2026 – May 2026

- Engineered a multi-agent system coordinating specialized agents for search and comparison; implemented **Model Context Protocol (MCP)** servers for secure real-time API access to shopping databases.
- Designed the data orchestration layer using LangGraph to manage complex state transitions and tool-calling dependencies across distributed nodes.

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, Data Ingestion

Dec 2024 – Apr 2025

- Built a multimodal **data ingestion pipeline** supporting PDFs, images, and YouTube transcripts; implemented semantic chunking and dense retrieval via FAISS/Pinecone for context-accurate QA.
- Optimized retrieval precision by designing efficient metadata schemas and vector index strategies for sub-second similarity search.

BulkyMail Automation | Python, Jinja2, CSV Processing

Dec 2024

- Developed a bulk email automation tool using Python and Jinja2 templating; engineered efficient CSV processing and dynamic data injection for large-scale personalized outreach.

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines, Data Modeling (Star/Snowflake), Data Ontology, Data Lineage, Master Data Management (MDM), Knowledge Graphs, Semantic Search

Languages & DBs: Python (Expert), SQL (Advanced), PostgreSQL, MySQL, FAISS, Pinecone, Apache Solr, Neo4j (Basic)

Orchestration & Tools: FastAPI, Airflow (Conceptual), AWS Glue (Exposure), Docker, Linux, Git, GitHub Actions, Jinja2

Libraries: PyTorch, Hugging Face, LangChain, Apache Spark (Exposure), Pandas, NumPy, Scikit-learn, Tesseract OCR

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana